Beyond Short Names: Emoji Keyword Retrieval across Models and a Bounded Late-Component Test

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Abstract

We use a prospectively specified two-arm design to distinguish cross-model lexical readability from a component-level causal account. Text language models receive emoji as Unicode token sequences. The observational arm crosses 55 emoji and 55 non-emoji Unicode symbols with raw, official-name, word-disjoint CLDR-keyword, and literal codepoint views. Across nine prespecified model–tokenizer configurations, the emoji keyword-minus-codepoint retrieval effect is positive in unrestricted and exact four-view token-count-matched analyses. Every configuration exceeds the registered 0.02 floor, and the four-model partial-conjunction primary passes ($p = 0.0027$). The symbol comparison recovers official names broadly and finds smaller, heterogeneous keyword effects. The passed observational criterion admits a causal arm in Llama 3.2 3B: rank-8 subspaces learned from 128 source emoji are tested at six late attention and MLP outputs on 192 held-out emoji. Engineering controls reproduce the baseline exactly. No primary site satisfies the joint necessity-and-sufficiency criterion under this single-context, low-baseline test. A separately specified follow-up adds interchange-optimized subspaces, residual-stream sites, 384 additional emoji in four contexts, and full-state swaps as positive controls. Both primary sites are uncalibratable under its registered criterion. Full swaps reproduce both baseline retrieval rates at block 27 and have asymmetric effects at blocks 21 and 24, suggesting recovery from unchanged glyph-position context. Together, the tests separate reproducible lexical retrieval from a rank-8 causal account that remains unsupported.

1 Introduction

Readers encounter emoji as compact visual objects. A text language model instead receives encoded Unicode scalars and tokenizer-dependent sequences. An official label may therefore be recoverable because it co-occurs with the character in standards, documentation, or training data. That observation alone cannot establish rendered-glyph perception, a human-like concept, or a localized semantic circuit.

Single-codepoint emoji nevertheless make the recoverability question unusually clean. Under the word-disjointness constraint below, the character shares no substring, morpheme, or ASCII-alphanumeric word with the keyword it must retrieve, so any observed relation must be carried by learned association. This provides a controlled testbed for how text-only models attach lexical relations to opaque tokens, and a concrete auditing target for systems that consume such tokens at scale.

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A prospectively specified design separates two narrower questions. The observational arm asks whether a raw atomic emoji retrieves a word-disjoint CLDR keyword better than its literal `U+...` spelling after exact token-count matching. Conditional on that result, the causal arms ask whether a low-rank late intervention identifies a representation that is both necessary and sufficient for the relation — an auditable distinction between broad lexical readability and simple mechanistic accounts, with no claim of visual perception or human-like understanding.

The study is organized around four safeguards and their corresponding results:

- Four text views and three disjoint item sets separate the raw Unicode surface, official English CLDR name, word-disjoint CLDR keyword, and literal codepoint spelling. The causal source and evaluation sets receive no observational-arm evaluation, although their surfaces may have appeared in earlier studies.
- The observational primary passes: 4/4 core configurations and 9/9 total configurations are positive in unrestricted and exact four-view token-count-matched analyses. The cross-model partial-conjunction value is 0.0027.
- A conditionally admitted causal experiment passes all zero-hook and restoration controls and finds no primary block-24 attention or MLP output that is jointly necessary and sufficient. A secondary block-27 MLP patch raises the composite endpoint through keyword gains and codepoint losses; its necessity criterion fails.
- A second admitted experiment learns rank-8 subspaces by interchange optimization and calibrates every site with full-interchange ceilings. Its registered verdict is uncalibratable at both primary residual sites; the full-state swaps reveal a depth-dependent asymmetry that motivates direct tests of glyph-position context.

The observational result extends beyond exact-short-name retrieval under the prespecified endpoint. Neither causal experiment supports the tested rank-8 readout-token account. The full-state controls motivate a more targeted question: can later attention recover the relation from unchanged context at the glyph positions?

2 Related work

Textual sources of emoji meaning. Emoji representations have been learned from descriptions [Eisner et al., 2016], linked to lexical sense inventories [Wijeratne et al., 2017], and induced from usage through emoji prediction [Felbo et al., 2017, Barbieri et al., 2018]. We use CLDR names and keywords [Unicode Consortium, 2026] as source-auditable endpoints and ask which item-wise textual relations are recoverable from a raw Unicode input; co-occurrence with these resources remains a candidate explanation for every positive result.

Probing and tokenizer controls. Probe accuracy can reflect the diagnostic model or its control task [Hewitt and Liang, 2019], and tokenizer choice can materially change model comparisons [Rust et al., 2021]. Our item-wise nearest-neighbor readout avoids a learned diagnostic classifier. Unrestricted and exact four-view token-count-matched tests address length while leaving token identity and tokenizer construction uncontrolled. The readout also differs from whole-representation similarity measures such as CKA [Kornblith et al., 2019]: it asks whether the correct item is the nearest partner across two text views.

Causal intervention. Conventional probing performance need not measure how information is used [Elazar et al., 2021]. Causal mediation, interchange interventions, and model editing instead test whether internal variables affect computation [Vig et al., 2020, Geiger et al., 2021, Meng et al., 2022]. Optimization-based alignment methods learn the intervention subspace itself, by maximizing interchange accuracy [Geiger et al., 2024, Wu et al., 2023] or through sparse dictionary features [Bricken et al., 2023, Cunningham et al., 2024]. Our first causal test fixes the subspace by a closed-form cross-covariance rule before any evaluation forward; the second learns it by interchange optimization under a prespecified protocol. Both require random-adjusted necessity and sufficiency on disjoint evaluation sets, using prespecified layers, ranks, and endpoints.

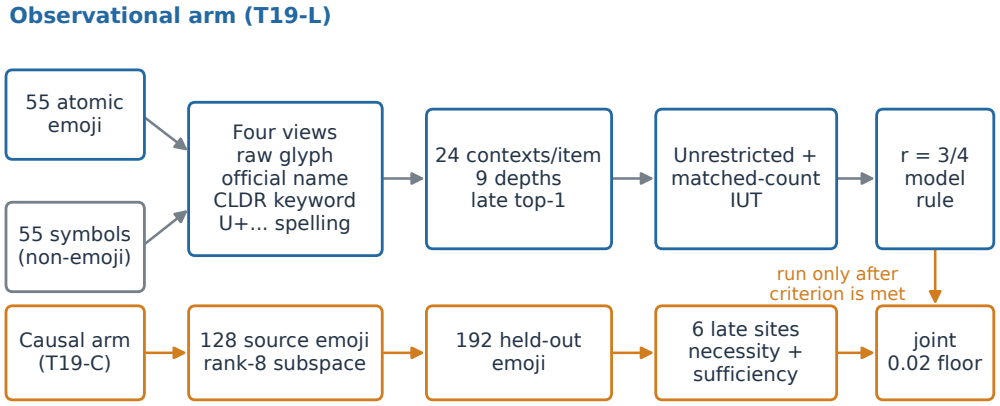


Figure 1: Prospective T19 design. T19-L crosses two glyph kinds with four text views. Only the emoji endpoint enters the registered cross-model primary. T19-C runs only after that criterion is met and uses disjoint source and evaluation emoji. No causal-set item receives a T19-L forward.

3 Prospective design

3.1 Input views and disjoint item sets

For auditability, we retain internal study labels: **T19-L** denotes the observational lexical-retrieval arm, **T19-C** the first conditionally admitted causal arm, and **T20-C** the second. No knowledge of the preceding project sequence is required; Appendix A gives the decision lineage without treating earlier exploration as confirmatory evidence.

Throughout, *glyph* means the raw Unicode scalar string supplied to a text model. No raster image is rendered. Atomic emoji are single-codepoint, fully-qualified Emoji 17 entries introduced no later than Emoji 13.1 [Unicode Consortium, 2025]. For each item, the official-name view is the exact English CLDR 48.2 `tts` value; the keyword view is one exact English CLDR `default` keyword. After lowercasing and extracting ASCII `[a-z0-9]+` words, official-name and keyword word sets must be nonempty and disjoint. The fourth view is the uppercase literal codepoint spelling. When several keywords qualify, one deterministic maximum bipartite matching assigns globally unique labels; a fixed SHA-256 edge order breaks ties before any T19 model forward. Table 1 in Appendix B shows concrete views for two emoji and one symbol.

The observational set contains 55 emoji and 55 non-emoji Unicode symbols. Symbols are single scalars in a Unicode general category beginning with `S`, have `Emoji=false`, have valid disjoint CLDR keywords, and are absent from every strict-UTF-8, no-NUL repository blob at the prespecified boundary. This freshness criterion concerns repository history. Exact emoji surfaces may have appeared in earlier experiments or pretraining; the prospective commitments cover the endpoint, keyword assignment, and item-set split.

Two further sets contain 128 causal-source and 192 causal-evaluation emoji. All 430 codepoints and 430 keywords are unique across sets. Figure 1 summarizes the separation and the one-way admission rule.

3.2 Observational readout

For each item and view, the prespecified evaluation pipeline appends one of two suffix carriers (`"a"` or `" |"`) at distances 1, 2, 4, and 8 and records the hidden state at the selected suffix token at nine normalized depths. Batch size is one; no chat template or added special token is used. The primary late window is fixed at depths 0.75, 0.875, and 1.0, yielding 24 contexts per item.

Within each view and context, hidden states are centered across items and row-L2-normalized; $z_i^{v,c}$ denotes the vector for item i, view v, and context c. Cross-view top-1 retrieval is

$$R_c(g \to v) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\left[i = \arg\max_j \left\langle z_i^{g,c}, z_j^{v,c} \right\rangle \right]. \tag{1}$$

The per-model primary estimate is

$$\Delta_m = \frac{1}{24} \sum_c \left[R_c(g \to \textsc{keyword}) - R_c(g \to \textsc{codepoint}) \right] \tag{2}$$

on the 55 observational emoji. This score measures nearest-neighbor retrieval from hidden states.

3.3 Token-count intersection and cross-model test

Every model must pass two components. The unrestricted component uses all 55 emoji and permutes keyword correspondence within glyph kind. The matched component retains non-singleton strata sharing the exact four-view token-count signature and permutes keywords only within stratum. Retrieval in that component uses the retained pool ($n = 31$–41 emoji across models), so its chance level is $1/n$; unrestricted and matched magnitudes are not treated as directly commensurable. Official-name and codepoint correspondences stay fixed. Thus the one-sided randomization test asks whether the fixed keyword mapping beats mappings allowed by its null; because the codepoint term is fixed, it is not a paired significance test of keyword–codepoint equality. The registered primary combines that test with the observed comparative requirement $\Delta_m \geq 0.02$ in both components. Each component uses 19,999 draws, and the model intersection–union test (IUT) p-value is the maximum of the two component values.

The sole cross-model primary uses four prespecified core configurations: Llama 3.2 3B base, Qwen2.5 7B base, OLMo 2 1B base, and Gemma 2 2B IT. It tests the partial conjunction that at least $r = 3$ of $m = 4$ configurations pass [Benjamini and Heller, 2008]. With ordered model IUT values $p_{(1)} \leq \cdots \leq p_{(4)}$, the registered value is $p_{\mathrm{PC}} = \min(1, 2p_{(3)})$. The other five configurations and all symbol or official-name results are secondary and cannot admit the causal arm.

Percentile intervals from 20,000 item-bootstrap replicates are reported as descriptive uncertainty, never as the primary decision.

3.4 Conditionally admitted causal test

After the observational criterion was committed, T19-C ran in the prespecified Llama 3.2 3B base configuration. The six sites are the attention and MLP component outputs at blocks 21, 24, and 27: the tensors returned by the attention output projection and by the MLP down-projection, hooked before their residual addition. At each site we learned a rank-8 basis from the left singular directions of the centered, row-normalized glyph–keyword cross-covariance on 128 source emoji. The control basis is constructed deterministically: sixteen seeded Gaussian directions inside the source span are deflated against the learned basis and orthonormalized, and the first eight are kept. Blocks 24 attention and MLP were primary; blocks 21 and 27 were secondary.

On 192 different evaluation emoji, the two criteria intervene on different forwards. Necessity intervenes on the glyph forward: with P the learned rank-8 projector and μ the source-set glyph mean at the site, the site output g is replaced by $g - P(g - \mu)$. Sufficiency intervenes on the codepoint forward: the site output c is replaced by $c + P(g - c)$, using the same item's glyph forward, with the replacement cast to BF16 at injection. Analogous operations in the random basis define adjusted effects:

$$N_s = E_s(\text{random removal}) - E_s(\text{learned removal}), \tag{3}$$
$$S_s = E_s(\text{learned patch}) - E_s(\text{random patch}). \tag{4}$$

Both source and evaluation inputs append eight copies of the single-token `" |"` suffix; the intervention and readout use its first token (distance 1). The endpoint reads the final hidden state of the intervened forward at that token as a query against two fixed 192-row galleries built once from clean,

un-intervened forwards: the keyword states and the codepoint states. Query matrix and each gallery are independently centered by their own item means, row-normalized, and scored by cosine top-1 identity, giving E_s = keyword top-1 minus codepoint top-1. The glyph query measures the baseline and necessity conditions. The patched codepoint query measures sufficiency; there the codepoint term is self-retrieval and equals 1 at baseline, so a patch can raise E_s either by real keyword gains or by destroying self-retrieval, and Section 4 reports the two components separately. E_s shares the item-wise contrast definition with T19-L while using a different item pool and one fixed context. A site must have both adjusted effects and both lower 95% item-bootstrap bounds above 0.02, pass both one-sided sign-flip tests (99,999 draws), pass zero-hook and remove-restore engineering controls, and survive Holm correction across the two primary sites [Holm, 1979]. No site, rank, depth, carrier, subset, or endpoint was replaced after observing the result.

3.5 Conditionally admitted optimized-subspace follow-up (T20-C)

The T19-C design left its own power uncalibrated. A second experiment, specified after the T19-C analysis was validated and recorded, makes five prespecified changes. (i) The rank-8 basis is learned by optimization: an orthogonally parametrized subspace is trained on the same 128 source emoji by interchange optimization (InfoNCE toward the own-item keyword state for the patched codepoint forward, plus a hinge pushing the ablated glyph forward below its own keyword), with a fixed 96/32 train/validation split, three seeds, and a fixed selection rule; the closed-form and random bases are recomputed at every site as comparators. (ii) The residual stream after blocks 21, 24, and 27 joins the six component sites, with the block-21 and block-24 residual sites primary. (iii) The decision uses the keyword component alone, with a codepoint self-retrieval guard (≥ 0.90 under the learned patch), so disruption cannot pass as sufficiency. (iv) Evaluation uses 384 atomic emoji with zero prior T19/T20 forwards. The original T19 selection had to be reproduced byte for byte before these items were drawn. Each item is read in four contexts (two carriers $\times$ distances 1 and 4), and its scores are averaged across contexts. (v) Every site has two full-state swap controls, termed *ceilings* in the protocol: the entire site state is exchanged between same-item glyph and codepoint forwards. By the registered rule, a primary site that fails while its sufficiency ceiling is below 0.06 is recorded as *uncalibratable at this endpoint*. This verdict leaves the low-rank account unresolved; the control cannot turn a failed decision positive. Floors, tests, Holm correction, and exact engineering checks are unchanged from T19-C.

3.6 Prospective specification and execution checks

Before acquisition, a checksum record identifies every protocol file after preliminary checks verify zero forward counts. An audit rechecks those files and requires the recorded version to have been pushed to the repository's default branch. Each causal arm starts only after a second record links this specification, the preceding validated analysis (T19-C for T20-C), and the item sets, with forward counts still zero. On each invocation, the execution program checks its own SHA-256 against this record and stops if it has changed. The anonymous supplement contains the dated specification and execution manifests (`*_freeze.json`, `*_go.json`), preliminary checks, and independent validation reports. Readers can therefore inspect the recorded order of specification, conditional admission, and evaluation.

4 Results

4.1 A lexical relation beyond the official short name

The registered cross-model primary passed. All four core configurations were positive; the third-smallest core IUT value was 0.00135, yielding $p_{\mathrm{PC}} = 2(0.00135) = 0.0027$. All nine prespecified configurations were positive and exceeded the registered 0.02 effect-size floor in both unrestricted and matched analyses (Figure 2; exact values in Appendix C). Model IUT values ranged from 0.00005 to 0.00185. The smallest unrestricted effect, Qwen2.5 0.5B at $+0.03636$, equals an average margin of two net top-1 matches per 55-item context. A validator that does not import the acquisition analyzer reconstructed all 36 randomization tests and reproduced the primary decision.

Across models, mean emoji keyword effects were $+0.0979$ unrestricted and $+0.1084$ matched. Mean official-name effects were larger, $+0.1440$ and $+0.1718$. Thus, retrieval extends beyond

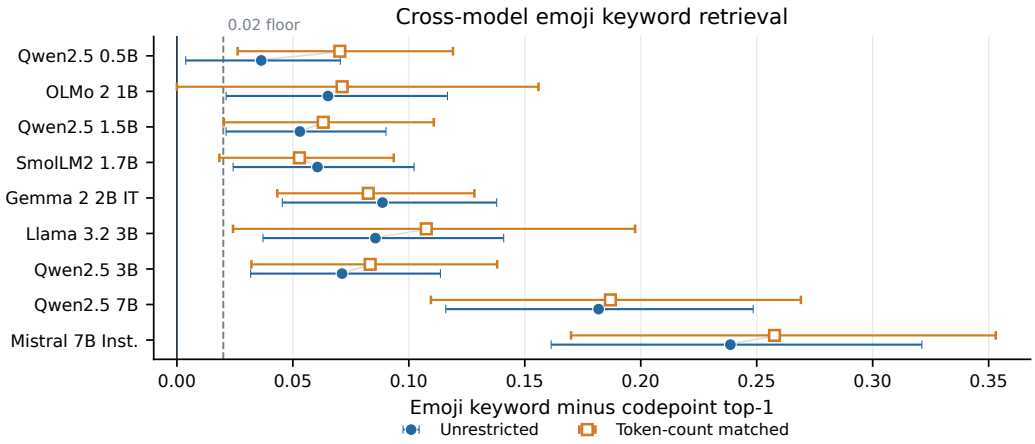


Figure 2: Emoji keyword-minus-codepoint top-1 effects in the registered roster order. Bars are descriptive 95% item-bootstrap intervals (20,000 replicates); randomization tests determine the primary decision. The 0.02 floor applies to point estimates. Unrestricted (U) and matched (M) analyses use different retained pools, so their magnitudes are not directly commensurable. Every displayed point estimate exceeds the floor.

the exact official name, although the official name remains the stronger key on average. The alternative label is an exact CLDR keyword, not a human-rated synonym or paraphrase; training-data co-occurrence could still fully explain the relation. Descriptively, the unrestricted effect also rises monotonically with parameter count inside the one family sampled at four sizes (Qwen2.5: $+0.036$, $+0.053$, $+0.071$, $+0.182$ from 0.5B to 7B). The roster remains confounded across families, so this ordering is description, not a registered scale result.

Depth was secondary. Across models and contexts, the unrestricted contrast was $+0.0409$ at depth 0 and $+0.0864$, $+0.1152$, $+0.0922$ at the three primary depths, whose mean is the $+0.0979$ above; onset and peak depths differed across configurations, so these data do not support one universal trajectory.

4.2 A named-symbol control limits the emoji-specific claim

Non-emoji symbols showed mean keyword effects of $+0.0236$ unrestricted and $+0.0153$ matched, compared with $+0.0979$ and $+0.1084$ for emoji. The emoji-minus-symbol point estimate was positive in all nine configurations in both components. Symbol keywords were heterogeneous: Qwen2.5 3B and 7B were negative in both components. Official-name retrieval exceeded keyword retrieval for symbols in 9/9 configurations in both components, with mean symbol official-name effects of $+0.0750$ and $+0.0821$ (Figure 3).

This comparison cannot establish an emoji-specific faculty: the panels are not matched on training frequency, lexical specificity, Unicode category, visual familiarity, or token identity. Its role is to prevent the incorrect inference that only emoji acquire readable names.

4.3 No prespecified late-component site passes both causal criteria

Before intervention, the final-hidden glyph endpoint on the 192-item causal evaluation set was $E_s = 0.07292$ (keyword top-1 0.07292, i.e. 14 of 192 items; codepoint top-1 0); the zero-hook engineering check reproduced this baseline identically at all six sites. This is the glyph-input baseline used by the necessity arm; the sufficiency arm starts from codepoint inputs, whose keyword top-1 is $1/192$ and whose codepoint term is self-retrieval at $192/192$. The 0.02 decision floor was therefore 27% of this baseline contrast, and the design contained no positive intervention control that calibrated attainable adjusted necessity at the primary sites. All six causal sites passed their engineering checks: zero-hook and remove-restore conditions reproduced the baseline final hidden state with exactly zero RMS difference. Bitwise equality is expected because baseline forwards

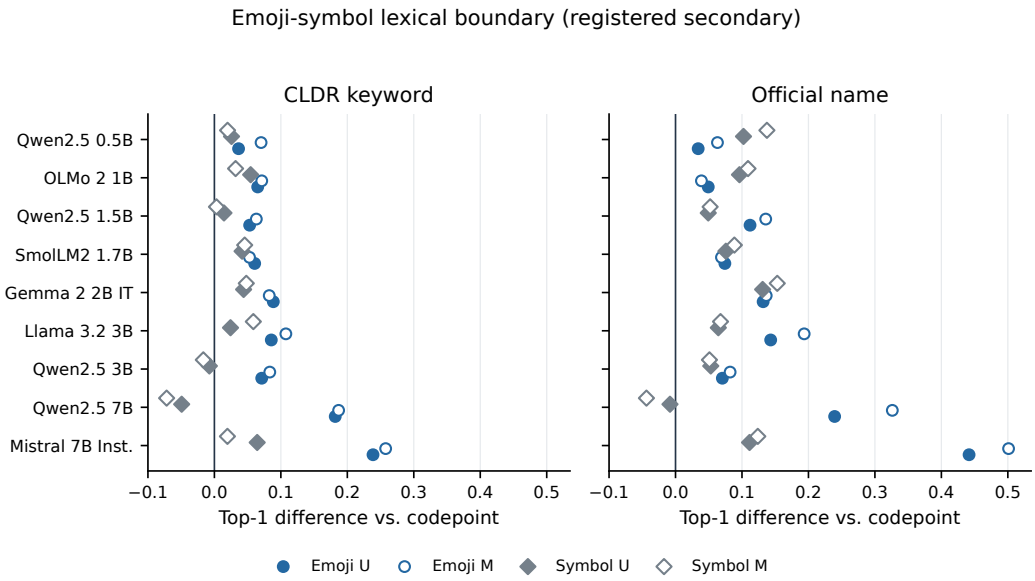


Figure 3: Registered secondary comparison. Each row shows unrestricted (U) and token-count-matched (M) effects for the same prespecified model. Official-name retrieval is not exclusive to emoji; disjoint-keyword effects are stronger and more consistent for the prespecified emoji panel. Glyph category, frequency, label quality, and token identity are not exchangeable, so this is a boundary result rather than an emoji-only semantic claim.

reuse the exact intervention batch geometry; any nonzero residue would indicate an instrumentation or batching mismatch. Neither primary block-24 site passed. Block-24 attention had adjusted necessity 0.0000 and sufficiency -0.00521; block-24 MLP had $+0.00521$ and 0.0000. Their site IUT values were 1.0 and 0.74912, respectively. None of the six sites passed both components (Figure 6, Appendix D; exact values in Table 4).

The secondary block-27 MLP patch produced adjusted sufficiency $+0.07812$ (95% interval $[+0.03646, +0.11979]$; one-sided $p = 0.00009$, uncorrected, with twelve such component tests run in total). Necessity was -0.01562 (interval $[-0.04167, +0.00521]$, $p = 0.96895$), giving a site IUT of 0.96895. The learned patch raised keyword top-1 from $1/192$ to $4/192$ while lowering codepoint self-retrieval from $192/192$ to $180/192$. The keyword gain accounts for 3 of the composite endpoint's 15 additional correct contrasts; the remaining 12 arise from lost codepoint self-retrieval. This pattern is consistent with broad representational disruption and does not identify a necessary keyword mechanism.

A post-hoc calibration, computed after the registered decision and labeled as such, replaced the entire component output with the same item's other-view output at each site (Appendix D). Full interchange at the primary block-24 sites moved keyword top-1 by 0.0000 (attention) and $+0.0052$ (MLP). These complete replacements fell below the 0.02 floor, limiting what the primary negative can establish about rank. The block-27 MLP composite ceiling ($+0.2448$ against a keyword-only $+0.0365$) quantifies how much of that endpoint's headroom was destroyed self-retrieval. These numbers motivated the registered ceilings of the follow-up.

4.4 Full-state controls reveal a depth-dependent asymmetry

Both primary T20-C residual sites return the registered verdict *uncalibratable at this endpoint* (site IUT values 0.624 and 0.679; sufficiency ceilings $+0.0234$ and $+0.0254$, below the registered 0.06 minimum). No secondary site passes either. All engineering checks are again exactly zero, and an independent validator re-derived every reported number, both verdicts, and the admission chain.

The registered full-state controls reveal a depth-dependent pattern (Figure 4; exact values in Appendix E). On the new evaluation set, glyph-input keyword top-1 is 0.0345 and codepoint-input

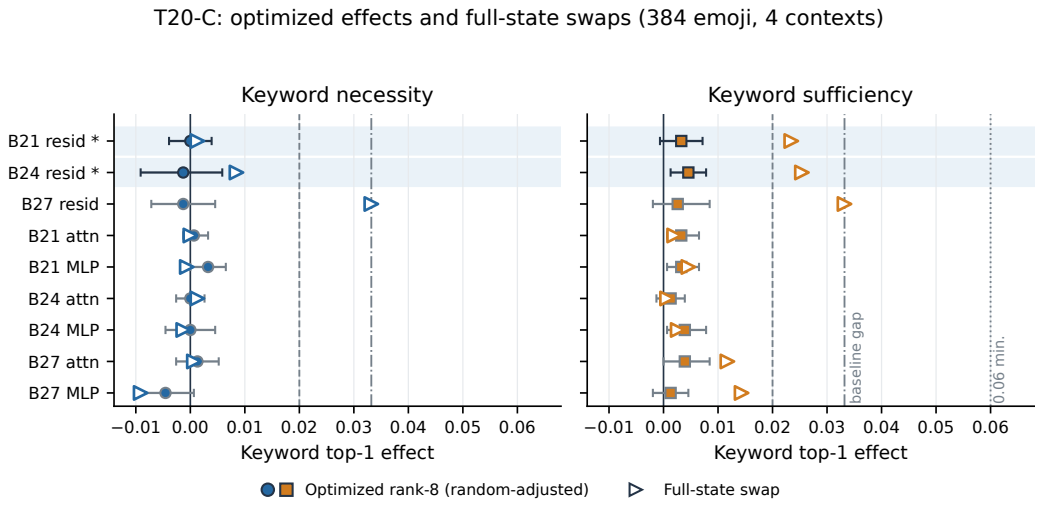


Figure 4: Registered T20-C outcomes over 384 fresh emoji in four contexts. Filled markers show random-adjusted rank-8 optimized effects with descriptive 95% item-bootstrap intervals; open triangles show the full-interchange ceilings at the same sites. Shaded rows mark the two primary residual sites; the dashed line is the registered 0.02 floor and the dotted line the 0.06 ceiling-informativeness minimum. Only the block-27 residual swap reproduces both baseline rates (the dash-dotted vertical line marks their 0.0332 gap). Neither primary sufficiency ceiling clears 0.06, so both sites are uncalibratable under the registered rule.

top-1 is 0.0013, a baseline gap of 0.0332. Swapping the readout-token residual stream at block 27 reproduces both baseline retrieval rates exactly. At blocks 21 and 24, injection transfers 71% and 76% of this gap; replacement in the glyph forward removes only 4% and 25%. Retrieval therefore remains near baseline after the readout-token stream is replaced at these earlier sites. Component-output swap effects at blocks 21 and 24 are smaller (at most 14% of the gap; block-24 attention 2%). This asymmetry suggests that later computation can recover the relation from unchanged context at the glyph positions, a pathway that remains to be tested directly.

The optimized subspaces fit source items unevenly: final InfoNCE is 0.52–0.65 at the block-21 residual against a 4.85 chance level, and 5.0–6.3 at attention outputs. Held-out source validation stays at zero to one item across all nine sites and three seeds. On the new evaluation set, the block-24 residual has adjusted sufficiency $+0.0046$ ($p = 0.0085$, uncorrected component test), less than one quarter of the 0.02 floor. The block-21 patch also fails the self-retrieval requirement ($0.818 < 0.90$). The source-item fit thus yields little held-out benefit under the tested rank-8 constructions.

5 Discussion

The arms answer different questions. Under a fixed text-only readout, T19-L recovers the same item across raw emoji and a word-disjoint CLDR keyword more reliably than across raw emoji and a literal codepoint string; exact token-count matching makes a pure length account insufficient, and the symbol panel places this relation within a broader pattern of named-symbol retrieval.

The causal arms expose two measurement constraints. T19-C's closed-form rank-8 account fails its joint criterion at sites where subsequent full-state controls produced small effects. T20-C registers these controls and returns uncalibratable at its primary sites. Its readout-token swaps also reveal an asymmetry: injection transfers much of the baseline gap at blocks 21 and 24, where replacement in the glyph forward leaves most retrieval intact. Both directions reproduce the baselines at block 27. One explanation is recovery from unchanged glyph-position context through later attention; a prospective attention-path intervention could test that explanation. The weak held-out effects of the optimized rank-8 subspaces leave this question open. T19-C's block-27 MLP patch illustrates a separate pitfall: most of its composite improvement comes from degrading codepoint self-retrieval. Reporting the endpoint's components separately exposes that source of apparent sufficiency.

6 Limitations and broader impact

Measurement. CLDR keywords provide standardized annotations without human ratings of synonymy, specificity, or familiarity in this study. Retrieval measures nearest-neighbor geometry at selected suffix tokens. Generation, task behavior, and rendered-glyph perception were not tested. Token-count matching does not match token identities or equalize tokenizer construction.

Generalization. The nine model–tokenizer configurations are a prespecified, convenience-constrained roster spanning several families and sizes [Yang et al., 2024, Groeneveld et al., 2024, Dubey et al., 2024, Gemma Team et al., 2024, Jiang et al., 2023, Ben Allal et al., 2025]. Family, parameter count, training data, tokenizer, and instruction tuning are confounded, so the study supports neither a scale law nor a model-population estimate. Emoji surfaces may have appeared in pretraining; the symbol-panel freshness criterion concerns only repository history.

Mechanism. Both causal results cover one Llama 3.2 3B checkpoint and rank 8, and both intervene only at the readout-token position of single sites; multi-position and attention-path interventions from the glyph tokens, sparse feature bases [Bricken et al., 2023, Cunningham et al., 2024], and nonlinear readouts were not evaluated. T19-C lacked a positive intervention control; T20-C registered full-interchange ceilings [cf. interchange alignment; Geiger et al., 2024, Wu et al., 2023] and classified its primary sites as uncalibratable. These swaps are reference interventions, not proven upper bounds on every subspace effect. Training on 96 items leaves item-specific coding, limited training data, and optimization limits as explanations for weak generalization. The baseline retrieval gap on the 384-item set (0.0332) is itself below the 0.06 calibration threshold chosen from the smaller T19-C set. Future designs should calibrate thresholds on a separate development set before specifying the evaluation protocol.

Reproducibility and impact. The anonymous compact archive binds configurations, selections, summaries, tests, validators, licenses, and the rendered paper; it omits raw arrays retained in the execution environment. Exact T19 acquisition duration and peak memory were not recorded in the submitted evidence. Better Unicode auditing could reveal hidden representation failures in multilingual systems; conversely, treating CLDR correspondence as a person's meaning, sentiment, culture, or identity could enable unjustified profiling, and the measurements should not be used for such inference. No personal data, human subjects, or user messages are used.

7 Conclusion

Across nine prespecified model–tokenizer configurations, raw atomic emoji retrieve word-disjoint CLDR keywords better than literal codepoints after exact four-view token-count matching, and a named-symbol comparison bounds the specificity of that observation. Two separately partitioned causal experiments leave the tested rank-8 readout-token account unsupported. Registered full-state controls reveal a depth-dependent asymmetry and motivate direct tests of recovery from glyph-position context. The results establish lexical readability across the specified models while identifying the calibration and pathway questions that a causal account must resolve.

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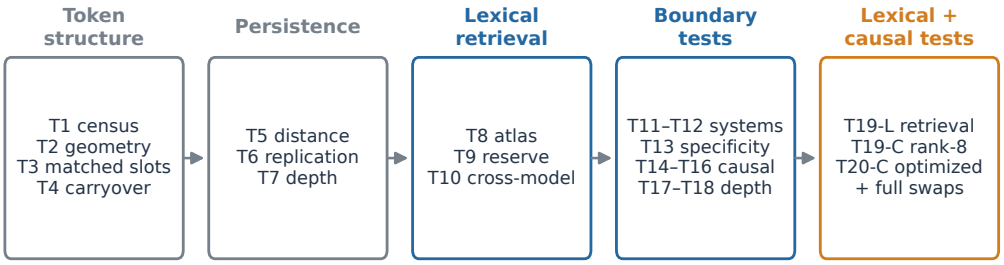


Figure 5: Exploration-to-confirmation lineage. Early studies exposed token-structure and persistence explanations; later studies narrowed the endpoint and causal test. Arrows record design decisions, not evidence pooling. Only the T19-L primary supports the confirmatory lexical claim in this paper.

A Exploration path and decision lineage

T19-L denotes the observational arm of the nineteenth study in this project. The concise lineage below includes only results that changed the next design decision. Earlier studies retain their exploratory or failed-confirmation status; the T19-L primary uses its own prespecified evidence alone.

Figure 5 groups the sequence by the question each stage addressed. The labels trace how the design changed.

Initial screen and matched controls (baseline and Milestone 2). A colored-shape steering screen produced 25 positive results among 36 layer–seed–dose configurations. An exact tokenization-matched follow-up supported one tested layer contrast and failed to support the other. We made tokenization part of the estimand, retained negative configurations, and stopped treating a pooled steering score as one stable semantic effect.

Token structure and short-range carryover (T1–T4). T1, the *multifamily tokenizer census*, found no universal token-count and prefix match for the fixed colored-shape panel. T2, *multifamily colored-shape geometry*, yielded no dual-model recurrence at the selected midpoint. T3, *five-family matched-slot geometry*, found recurrence only in OLMo, where the exact shared slot token was already a sufficient explanation. T4, *common-suffix carryover*, showed that the OLMo pattern persisted one common token later. The sequence converted tokenizer structure from a nuisance variable into an explicit competing explanation.

Persistence, depth, and the lexical turn (T5–T8). T5, the *carrier-distance profile*, and T6, the *dense CUDA carrier-distance replication*, both failed their prespecified distance-attenuation conjunctions. T7, *Mistral base depth-by-distance*, passed an early-versus-late interaction in a model and endpoint selected after T6, so it remained prospective within that configuration rather than independent confirmation. T8, the *Mistral semantic atlas*, then mapped 842 discovery emoji and found a large descriptive name-over-codepoint contrast. That atlas supplied the endpoint later tested on the untouched T9 reserve.

Lexical exploration and failed reserves (T9–T12). A held-out Mistral configuration first supported name-over-codepoint retrieval. A nine-configuration follow-up retained statistical signal and failed its registered cross-model effect-size rule. Broader glyph-system exploration then suggested Unicode-composition hypotheses; neither survived testing on separately reserved items. We kept both failures, dropped the generic composition claim, and returned to a narrower lexical question.

Specificity decomposition and first causal localization (T13–T14). T13 separated structured source directions from generic target susceptibility on already exposed panels and remained ex-

ploratory. T14 then paired a signed six-model intervention grid with a separately specified selected-site test. The broad response pattern was heterogeneous across model, depth, source form, null family, and sign; the selected Llama component failed the joint causal criterion on a new evaluation set. This combination motivated the response-shape tests and stricter causal separation in T15–T16.

Readable responses and unsuccessful causal tests (T15–T16). An emoji–word response pattern replicated prospectively, and prespecified forced-choice associations were detectable. Yet a subspace intervention and a multi-layer transplantation each failed their stronger causal criteria. This separated readout from implementation and motivated the present joint necessity–sufficiency criterion.

Depth mapping and fresh confirmation (T17–T18). Across 292 previously exposed atomic emoji, name-minus-codepoint retrieval was positive in all nine model configurations at middle-to-late depths. Peaks were model-specific, and final item retrieval (9/9) did not require consistent whole-representation CKA alignment (4/9 configurations). We therefore prespecified a common late window, replaced the population, and made no shared-transition or universal-manifold claim. On 192 repository-fresh composite emoji, the official-name contrast then passed in 4/4 core configurations and 8/9 configurations overall after exact token-count matching. This justified asking whether the relation extends beyond the official short name, while adding a named non-emoji control and reserving new item sets for causality. This T18 composite set is distinct from the 192 atomic emoji in the T19-C evaluation set.

T19-L and T19-C (confirmatory and first conditional arm). T19-L replaces the sole label endpoint with a word-disjoint CLDR keyword and adds literal-codepoint and named-symbol boundaries. T19-C tests the prespecified rank-8 component account on disjoint source and evaluation sets. Broad lexical retrieval passes, while the simple late-component mediator fails its joint criterion.

T20-C (second conditional arm). The T19-C result, recorded without a positive control, motivated a separate prespecified experiment. T20-C was specified and authorized after the T19-C analysis was validated; its execution manifest identifies that analysis by checksum. Its five changes (optimized subspaces, residual-stream sites, keyword-only decision, a new 384-item four-context evaluation set, and registered full-state controls) each address a T19-C limitation. The registered verdict is uncalibratable at both primary sites. The measured swap asymmetry motivates the pathway hypothesis in Section 4.4.

This lineage explains why the final design contains four text views, exact token-count strata, a non-emoji symbol panel, three disjoint item sets, and a necessity–sufficiency intersection. It also explains what is absent: no favorable prior layer, glyph system, or causal site was promoted into the T19-L primary. Only that prespecified primary supports the paper's confirmatory lexical claim.

B Prespecified examples and model roster

Raw scalar†	CLDR `tts`	Keyword	Literal view
U+1F923	rolling on the floor laughing	lmao	`U+1F923`
U+1F928	face with raised eyebrow	skeptic	`U+1F928`
U+00B4	acute accent	tone	`U+00B4`

Table 1: Concrete prespecified views for two emoji and one non-emoji symbol. †The model receives the scalar itself; the table names it by codepoint because the paper typeface need not render every glyph. The literal view is the displayed ASCII string.

Table 2 reports the exact model artifacts and revisions used by the prespecified observational roster. The Llama configuration uses the BF16 mirror artifact specified by the protocol. Despite its `mlx-community` identifier, the recorded run loaded its safetensors through the Transformers/CUDA backend; we make no upstream bit-identity claim. Model cards and upstream license terms are part of the submission-archive license audit. The cards declare Apache-2.0 for the Qwen2.5 0.5B, 1.5B, and 7B configurations, OLMo, SmolLM2, and Mistral; Qwen2.5 3B uses the non-commercial Qwen

Research License, Gemma uses the Gemma Terms of Use, and the Llama mirror uses the Llama 3.2 Community License. No checkpoint or tokenizer file is redistributed.

Role	Artifact identifier	Revision
Qwen2.5 0.5B base	`Qwen/Qwen2.5-0.5B`	`060db6499f32`
OLMo 2 1B base	`allenai/OLMo-2-0425-1B`	`a1847dff3500`
Qwen2.5 1.5B base	`Qwen/Qwen2.5-1.5B`	`8faed761d45a`
SmolLM2 1.7B base	`HuggingFaceTB/SmolLM2-1.7B`	`effd688a1292`
Gemma 2 2B IT	`google/gemma-2-2b-it`	`299a8560bedf`
Llama 3.2 3B base	`mlx-community/Llama-3.2-3B-bf16`	`60a99aaf4316`
Qwen2.5 3B base	`Qwen/Qwen2.5-3B`	`3aab1f1954e9`
Qwen2.5 7B base	`Qwen/Qwen2.5-7B`	`d14972939875`
Mistral 7B Inst. v0.3	`mistralai/Mistral-7B-Instruct-v0.3`	`c170c708c41d`

Table 2: Model artifact identifiers and the first 12 hexadecimal characters of the exact revisions used. Full revisions are stored in the submission manifest.

C Exact observational results

Model configuration	U effect	M effect	Model IUT p	Pass
Qwen2.5 0.5B	0.0364	0.0702	0.00055	yes
OLMo 2 1B	0.0652	0.0712	0.00135	yes
Qwen2.5 1.5B	0.0530	0.0631	0.00185	yes
SmolLM2 1.7B	0.0606	0.0528	0.00005	yes
Gemma 2 2B IT	0.0886	0.0825	0.00005	yes
Llama 3.2 3B	0.0856	0.1075	0.00135	yes
Qwen2.5 3B	0.0712	0.0833	0.00005	yes
Qwen2.5 7B	0.1818	0.1869	0.00005	yes
Mistral 7B Inst.	0.2386	0.2577	0.00005	yes

Table 3: Exact reporting values for the registered emoji keyword-minus-codepoint endpoint. U is unrestricted; M is exact four-view token-count matched. The model IUT is the maximum component randomization p-value.

Matched emoji counts ranged from 31 to 41. The matched component retained at least 63 observational items total in every configuration, with at least 31 emoji and 32 symbols; the unrestricted component used all 55 emoji and 55 symbols. The acquisition comprised $9 \times 110 \times 4 \times 2 = 7{,}920$ batch-one forwards (configurations $\times$ items $\times$ views $\times$ carriers). Four distances and nine depths produced $7{,}920 \times 4 \times 9 = 285{,}120$ readout vectors in 45 checksum-verified data files.

D Exact causal results

Block	Site	Primary	N_s	p_N	S_s	p_S	Site IUT
21	attention	no	-0.00521	1.00000	0.00521	0.49983	1.00000
21	MLP	no	0.00000	1.00000	0.00521	0.49760	1.00000
24	attention	yes	0.00000	1.00000	-0.00521	1.00000	1.00000
24	MLP	yes	0.00521	0.49989	0.00000	0.74912	0.74912
27	attention	no	0.00000	1.00000	0.01042	0.31511	1.00000
27	MLP	no	-0.01562	0.96895	0.07812	0.00009	0.96895

Table 4: Random-adjusted necessity (N_s) and sufficiency (S_s). All six engineering checks passed. Holm correction applies across the two primary sites and did not change either decision; no site passed the component-level intersection.

The causal run used $128 \times 2 = 256$ source forwards. For each of 192 evaluation items, three baseline-view forwards plus two batched intervention forwards at each of six sites yielded $192 \times 15 = 2{,}880$ evaluation forwards; seven intervention rows per site yielded $192 \times 6 \times 7 = 8{,}064$ rows. Each of 12 component tests used 99,999 sign flips and 20,000 item-bootstrap replicates. An independent validator rehashed source and evaluation arrays, reconstructed all 12 tests and intervals, verified

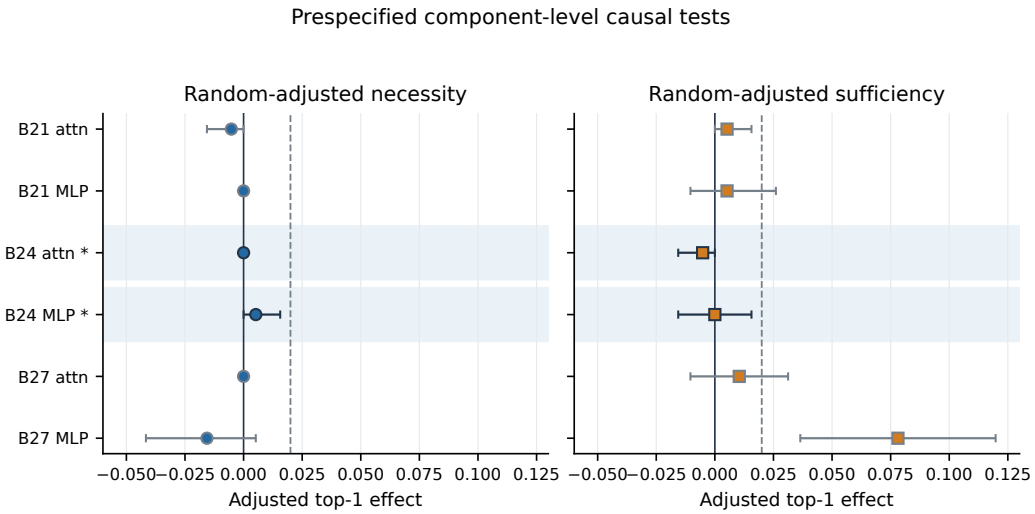


Figure 6: Random-adjusted T19-C causal effects with descriptive 95% item-bootstrap intervals over 192 emoji held out from T19-L. Shaded rows and asterisks mark the two primary block-24 sites. Solid vertical lines show zero; dashed lines show the registered 0.02 effect-size floor. No site has both necessity and sufficiency above the floor with positive lower bounds.

learned and random subspace relations, and reproduced six engineering passes and zero positive sites.

Block	Site	Necessity ceiling	Sufficiency ceiling	Composite suff. ceiling
21	attention	+0.0052	0.0000	0.0000
21	MLP	0.0000	+0.0156	+0.0156
24	attention	+0.0052	0.0000	0.0000
24	MLP	−0.0156	+0.0052	+0.0156
27	attention	−0.0156	+0.0417	+0.0417
27	MLP	−0.0208	+0.0365	+0.2448

Table 5: Post-hoc full-interchange ceilings at the six T19-C sites on the 192-item set and single registered context (keyword top-1 unless marked composite). Computed after the registered decision from 2,304 additional forwards; labeled descriptive and unable to change the recorded outcome. Both primary block-24 sufficiency ceilings sit at or near zero, far below the registered 0.02 floor.

E Exact optimized-subspace follow-up results

Table 6 reports the registered T20-C quantities at all nine sites: random-adjusted effects of the optimized rank-8 subspace on the keyword component, the full-interchange ceilings, the codepoint self-retrieval guard under the learned patch, and the site IUT value. The glyph-input baseline on the 384-item set is keyword top-1 0.03451 (codepoint top-1 0.00195); the codepoint-input baseline is keyword top-1 0.00130 with self-retrieval 1.0, giving a baseline keyword-retrieval gap of 0.03320. The closed-form T19 construction, recomputed at the same sites as a registered secondary, reaches at most +0.0117 adjusted sufficiency (block-27 residual) and never exceeds the floor. All engineering RMS values are exactly zero. The training stage used 384 capture forwards and 33,048 batched training passes on source items only; evaluation used 32,256 forwards and 124,416 intervention rows; 18 component tests used 99,999 sign flips and 20,000 bootstrap replicates each. The independent validator re-derived every value in Table 6, both primary verdicts, and the specification and execution records.

Subspace training used Adam with learning rate 10^{-3} for 600 steps, 32 items per batch, and seeds 0, 1, and 2. The 128 source items were split deterministically into 96 training and 32 validation items. Seed selection maximized validation keyword top-1 under patching minus that under ablation; ties

selected the lowest seed. Training used the pipe carrier at distance 1, FP32 subspace parameters, and BF16 injection during evaluation. The model weights remained unchanged.

Block	Site	Primary	N_s^{opt}	S_s^{opt}	Nec. ceiling	Suff. ceiling	Guard	Site IUT	Verdict
21	residual	yes	0.0000	$+0.0033$	$+0.0013$	$+0.0234$	0.818	0.624	uncalibratable
24	residual	yes	-0.0013	$+0.0046$	$+0.0085$	$+0.0254$	0.971	0.679	uncalibratable
27	residual	no	-0.0013	$+0.0026$	$+0.0332$	$+0.0332$	0.963	0.745	—
21	attention	no	$+0.0007$	$+0.0033$	0.0000	$+0.0020$	0.999	0.499	—
21	MLP	no	$+0.0033$	$+0.0033$	-0.0007	$+0.0046$	0.972	0.065	—
24	attention	no	0.0000	$+0.0013$	$+0.0013$	$+0.0007$	1.000	0.687	—
24	MLP	no	0.0000	$+0.0039$	-0.0013	$+0.0026$	0.987	0.612	—
27	attention	no	$+0.0013$	$+0.0039$	$+0.0007$	$+0.0117$	0.990	0.377	—
27	MLP	no	-0.0046	$+0.0013$	-0.0091	$+0.0143$	0.971	0.977	—

Table 6: Registered T20-C results. $N_s^{\mathrm{opt}}/S_s^{\mathrm{opt}}$ are random-adjusted keyword-component effects of the optimized rank-8 subspace, averaged over four contexts on 384 fresh emoji. Ceilings replace the entire site state with the same item's other-view state. Guard is codepoint self-retrieval under the learned patch (floor 0.90). Verdicts apply to primary sites under the registered rule; the block-27 residual swap effects equal the baseline keyword-retrieval gap 0.0332 exactly.

F Execution, licensing, and artifact boundary

Both observational and causal acquisition used a fixed Transformers/CUDA BF16 environment and an NVIDIA GeForce RTX 5090. Model artifacts, tokenizer replay receipts, selection files, analysis summaries, randomization seeds, bootstrap seeds, and validators are checksum-bound. Figure inputs in the paper are only the compact validated model-configuration and causal-site JSON summaries; a generated manifest hashes both inputs and every plotted PDF/PNG output.

The anonymous evidence archive is a deterministic whitelist-only zip. Its manifest records every payload path, byte count, and SHA-256; a validator rejects unsafe paths, author or local-path tokens, model-weight formats, and raw array formats. The archive includes the paper, compact results, selected item sets, code, validation receipts, and paired license/compute/ethics reports. It omits model-dependent activation arrays, learned causal subspaces, the full intervention ledger, model weights, and tokenizer files. Those raw scientific objects remain identified by checksums in execution records. The package supports compact-summary and figure audit; replay from raw arrays requires the omitted files.

The observational acquisition used 7,920 batch-one forwards and produced 285,120 readout vectors. T19-C acquisition used 256 source forwards, 2,880 evaluation forwards, and 8,064 intervention rows, plus 2,304 post-hoc ceiling forwards. T20-C used 384 capture forwards, 33,048 batched training passes (gradients reach only the 3072×8 subspace parameters; no model weight is trained), 32,256 evaluation forwards, and 124,416 intervention rows, with recorded stage durations of 14.7 and 7.3 minutes. No generation was performed anywhere. Exact wall-clock time and peak memory for the T19 stages are absent from the submitted records; the compute checklist answer reflects that gap.

Selected Unicode 17 and CLDR 48.2 rows are used under Unicode License v3 and travel with its copyright and permission notice. New project code is Apache-2.0. Checkpoint assets are identified by exact revision and upstream terms; the archive omits their files. In particular, the Qwen2.5 3B configuration is used only for non-commercial research under the Qwen Research License.

We reviewed the study against the NeurIPS Code of Ethics. It uses no human participants, personal data, private messages, or scraped user content. LLM coding and research assistants supported protocol drafting, code generation, manuscript editing, and one author-side fresh-eyes review; they did not provide labels, choose outcomes after T19 execution, or serve as the scientific evaluator. The authors retained execution authority and approved the prespecified endpoints and final claims; numerical evidence and validators were checksum-bound.

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   Answer: [Yes]

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   Question: Does the paper discuss the limitations of the work performed by the authors?

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   Justification: Section 6 discusses measurement, tokenization, model-roster, causal-scope, archive, and social-interpretation limits.

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5. **Open access to data and code**

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6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

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   Justification: Section 3 specifies the evaluation sets, carriers, distances, depths, readouts, tests, interventions, and sites. Section 3.5 gives the T20-C objective; Appendix E supplies the optimizer, training split, and seed-selection rule. No model weights are trained.

7. **Experiment statistical significance**

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    Justification: Appendices B and F and the archive license register identify every model and data term. Model/tokenizer files are not redistributed; the Qwen 3B configuration is bounded to non-commercial research; and the Unicode License v3 notice accompanies the derived rows.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The anonymous artifact includes paired artifact documentation, a machine-readable file manifest, SHA256SUMS, omission boundaries, validation instructions, licenses, prespecified selections, compact evidence, and source code.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The study uses no crowdsourcing or human subjects.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The study uses no crowdsourcing, personal data, or human subjects.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: Appendix F discloses assistant support for protocol drafting, code generation, editing, and fresh-eyes review, while separating it from labels, prespecified decisions, execution authority, numerical evidence, and scientific evaluation.